

Inheritance Language: Allele to Phenotype

LAUNCH • Science • 40 minutes

I can use inheritance vocabulary to connect alleles, genotype and phenotype in a simple monohybrid model.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
allele	version of a gene
genotype	allele-letter combination
phenotype	observed characteristic

First idea

In a simplified pea model, P is a purple-flower allele and p is a white-flower allele. Which item is a genotype?

- A. Pp
- B. purple flowers
- C. the whole pea plant

My first idea and reason:

flower-colour gene: alleles P (purple) and p (white)

two alleles per plant, one from each parent

Gene and alleles

We do • choose, explain, check

Which statement is correct for the simplified P/p model?

- A. Pp is a heterozygous genotype with the purple phenotype.
- B. P is dominant because purple is biologically better.
- C. pp is heterozygous because the letters are lowercase.

My choice: _____ Evidence or reason:

Genotype-phenotype decoder

Classify each card, then justify the phenotype using the simplified dominance rule.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
HOMOZYGOUS	
HETEROZYGOUS	
PHENOTYPE STATEMENT	

Activity cards

1 • PP	2 • Pp
Two copies of the dominant allele.	One dominant and one recessive allele.
3 • White flowers	
The observable characteristic produced by pp in this model.	

Use genotype, phenotype and homozygous/heterozygous in one accurate explanation.

Your lesson record

Decode six genotypes, identify phenotypes and explain the limits of the simple P/p model.

Model helps us see

Letter tiles make allele combinations and genotype-to-phenotype mapping visible before probability is introduced.

Model does not show

The model assumes one gene, two alleles and complete dominance; many human and organism characteristics are more complex and influenced by environment.

Supported

Match PP, Pp and pp to same/different alleles and purple/white phenotype.

Keep the P/p key and icon-backed word mat visible.

Standard

Classify six genotype/phenotype statements and explain two matches.

Use: ___ is the genotype; ___ is the phenotype because ___.

Stretch

Correct three dominance misconceptions and evaluate the one-gene complete-dominance model.

Do not introduce codominance, sex linkage or Higher-tier drift.

Evidence target

Accurate use of allele, genotype, phenotype, homozygous and heterozygous plus one model limitation.

Exit, Audience and evidence • W13L1

Supported: Genotype or phenotype: Pp?

Standard: Explain why Pp is heterozygous and gives purple flowers in this model.

Stretch: Why does dominant not mean better or more common?

What I said, and what it changed

Next I will _____.

Adult witness note

What was observed, and with what level of independence?

My evidence and explanation

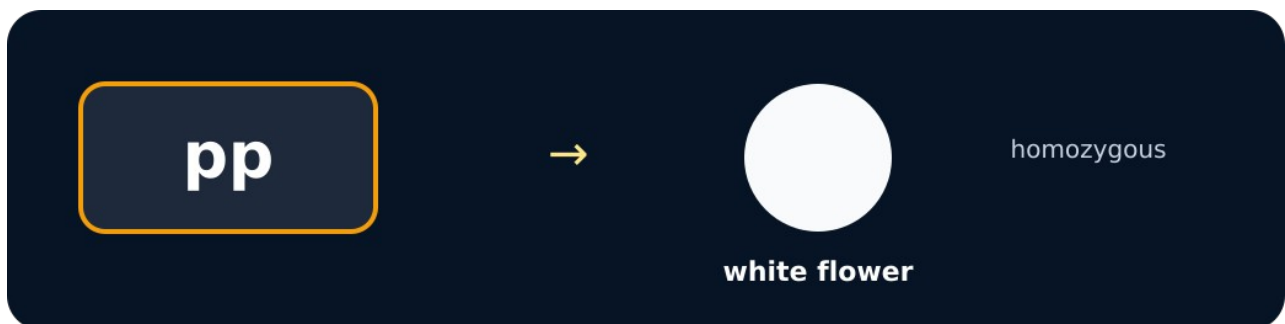
What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



Pp genotype



pp genotype

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
